Quadratics & Nonlinear Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Factoring Quadratic Equations

Factor each quadratic equation completely and find all real roots:

a $2x^2 - 11x + 12 = 0$ $(2x - 3)(x - 4) = 0 \Rightarrow x = \frac{3}{2}$ or $x = 4$

b $9x^2 - 25 = 0$ $(3x - 5)(3x + 5) = 0 \Rightarrow x = \frac{5}{3}$ or $x = -\frac{5}{3}$

Use AC-method factoring or the difference-of-squares identity.

2 [Foundational] • [FRQ] • Target: Completing the Square & Vertex Form

Rewrite $f(x) = 2x^2 - 12x + 11$ in vertex form $f(x) = a(x - h)^2 + k$ by completing the square, and state the coordinates of its vertex and the equation of its axis of symmetry.

Factor 2 from the first two terms: $f(x) = 2(x^2 - 6x + 9) + 11 - 18 = 2(x - 3)^2 - 7$. The vertex is $(3, -7)$ and the axis of symmetry is the vertical line $x = 3$.

3 [Foundational] • [FRQ] • Target: Quadratic Formula & Simplifying Radicals

Use the quadratic formula to find the exact solutions of $3x^2 + 4x - 5 = 0$ in simplest radical form.

$x = \frac{-4 \pm \sqrt{4^2 - 4(3)(-5)}}{2(3)} = \frac{-4 \pm \sqrt{16 + 60}}{6} = \frac{-4 \pm \sqrt{76}}{6} = \frac{-4 \pm 2\sqrt{19}}{6} = \frac{-2 \pm \sqrt{19}}{3}$.

4 [Foundational] • [FRQ] • Target: Discriminant & Nature of Roots

Compute the discriminant $\Delta = b^2 - 4ac$ for each quadratic equation and state the number of distinct real solutions:

a $4x^2 - 12x + 9 = 0$ $\Delta = (-12)^2 - 4(4)(9) = 0$ (1 repeated real solution, $x = \frac{3}{2}$)

b $2x^2 - 3x + 5 = 0$ $\Delta = (-3)^2 - 4(2)(5) = -31 < 0$ (0 real solutions)

A quadratic has 2 distinct real roots if $\Delta > 0$, 1 repeated real root if $\Delta = 0$, and 0 real roots if $\Delta < 0$.

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5 [Foundational] • [FRQ] • Target: Factored Form & Symmetry of Parabolas

A parabola has x-intercepts at $(-2, 0)$ and $(6, 0)$ and passes through $(0, -12)$. Write the equation of the parabola in factored form and standard form, and find its minimum value.

In factored form, $y = a(x + 2)(x - 6)$. Substituting $(0, -12)$ gives $-12 = a(2)(-6) \Rightarrow a = 1$. Thus $y = (x + 2)(x - 6) = x^2 - 4x - 12$. The vertex occurs at the midpoint $x = \frac{-2+6}{2} = 2$, giving minimum value $y(2) = (4)(-4) = -16$.

6 [Foundational] • [FRQ] • Target: Vieta's Formulas for Sum & Product of Roots

Without solving for the roots r_1 and r_2 of $5x^2 - 15x - 9 = 0$, use Vieta's formulas to find (a) $r_1 + r_2$, (b) $r_1 r_2$, and (c) $r_1^2 + r_2^2$.

By Vieta's formulas: (a) $r_1 + r_2 = -\frac{-15}{5} = 3$; (b) $r_1 r_2 = -\frac{9}{5}$; (c) $r_1^2 + r_2^2 = (r_1 + r_2)^2 - 2r_1 r_2 = 3^2 - 2\left(-\frac{9}{5}\right) = 9 + \frac{18}{5} = \frac{63}{5} = 12.6$.

7 [Foundational] • [FRQ] • Target: Projectile Motion Quadratic Modeling

A flare is launched vertically upward from a platform 24 m above the ground with initial velocity 32 m/s, so its height (in meters) after t seconds is $h(t) = -4.9t^2 + 29.4t + 24.5$. Find (a) the time t at which the flare reaches its maximum height, (b) the maximum height, and (c) the time when it hits the ground.

(a) $t_{\text{vertex}} = -\frac{29.4}{2(-4.9)} = 3$ seconds. (b) $h(3) = -4.9(9) + 29.4(3) + 24.5 = -44.1 + 88.2 + 24.5 = 68.6\text{ m}$. (c) Set $h(t) = 0$: divide by -4.9 to get $t^2 - 6t - 5 = 0 \Rightarrow t = 3 + \sqrt{14} \approx 6.74$ seconds.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Discriminant Condition for Tangency

For what positive value of k does the quadratic equation $3x^2 - kx + 12 = 0$ have a single repeated real solution, and what is that solution?

Set the discriminant to zero: $\Delta = (-k)^2 - 4(3)(12) = k^2 - 144 = 0$. Since $k > 0$, $k = 12$. Substituting $k = 12$ gives $3(x^2 - 4x + 4) = 3(x - 2)^2 = 0$, so the repeated root is $x = 2$.

9 [Intermediate] • [FRQ] • Target: Equations Quadratic in Form

Solve the bi-quadratic equation $x^4 - 13x^2 + 36 = 0$ for all real solutions x.

Let $u = x^2$: $u^2 - 13u + 36 = 0 \Rightarrow (u - 4)(u - 9) = 0$, so $x^2 = 4$ or $x^2 = 9$. Taking square roots yields four real solutions: $x \in \{-3, -2, 2, 3\}$.

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10 **[Intermediate]** • **[FRQ]** • **Target: Geometric Area Quadratic Modeling**

A rectangular garden of dimensions $12\,\text{m}$ by $20\,\text{m}$ is surrounded by a gravel walkway of uniform width w meters. If the total combined area of the garden and walkway is $384\,\text{m}^2$, find the width w of the walkway.

The outer dimensions are $(12+2w)$ and $(20+2w)$. Set $(12+2w)(20+2w)=384 \Rightarrow 4w^2+64w+240=384 \Rightarrow 4w^2+64w-144=0 \Rightarrow w^2+16w-36=0 \Rightarrow (w+18)$. Since $w>0$, $w=2\,\text{m}$.

11 **[Intermediate]** • **[FRQ]** • **Target: Revenue Optimization via Quadratic Vertex**

A theater currently sells 300 tickets per night at 20 dollars per ticket. Market research shows that for every 1-dollar increase in ticket price, nightly attendance drops by 10 tickets. Express nightly revenue $R(x)$ as a quadratic function of the number of 1-dollar price increases x, and find the ticket price that maximizes revenue.

Revenue is $R(x)=(20+x)(300-10x)=-10x^2+100x+6000$. The vertex occurs at $x=-\frac{100}{2(-10)}=5$. Thus a 5-dollar increase (to a ticket price of 25 dollars) maximizes revenue at $R(5)=25\times 250=6{,}250$ dollars.

12 **[Intermediate]** • **[FRQ]** • **Target: Parabola-Line Intersection**

The parabola $y=-x^2+4x+5$ and the line $y=x+1$ intersect at two points A and B, as shown. Find the exact coordinates of A and B.

Set $-x^2+4x+5=x+1 \Rightarrow x^2-3x-4=0 \Rightarrow (x-4)(x+1)=0$. For $x=-1$, $y=0$; for $x=4$, $y=5$. Thus $A=(-1,0)$ and $B=(4,5)$.

13 **[Intermediate]** • **[FRQ]** • **Target: Root Difference & Vieta's Relations**

If the roots of $x^2-6x+c=0$ differ by 4, find the value of the constant c and both roots.

Let the roots be r_1 and r_2. By Vieta's formulas, $r_1+r_2=6$. Since $r_1-r_2=4$, adding gives $2r_1=10 \Rightarrow r_1=5$ and $r_2=1$. Therefore $c=r_1r_2=5(1)=5$.

14 **[Intermediate]** • **[FRQ]** • **Target: Solving Quadratic Inequalities**

Solve the quadratic inequality $2x^2-7x-15\le 0$ and express the solution set in interval notation.

Factor the quadratic: $(2x+3)(x-5)\le 0$. The boundary roots are $x=-\frac{3}{2}$ and $x=5$. Since the parabola opens upward ($a=2>0$), the expression is non-positive on the closed interval $\left[-\frac{3}{2},5\right]$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • [FRQ] • Target: Symmetric Functions of Quadratic Roots • Cross-Lesson: Topic 10.01 (Rational Expressions & Vieta's Theorem)**

Let r and s be the roots of $2x^2 - 8x + 5 = 0$. Without solving for r and s individually, find the monic quadratic equation whose roots are $\frac{1}{r}$ and $\frac{1}{s}$, and compute $\frac{r}{s} + \frac{s}{r}$.

By Vieta's formulas, $r + s = 4$ and $rs = \frac{5}{2}$. For the reciprocal roots, $\frac{1}{r} + \frac{1}{s} = \frac{r+s}{rs} = \frac{4}{5/2} = \frac{8}{5}$ and $\frac{1}{r} \cdot \frac{1}{s} = \frac{2}{5}$, so the monic quadratic is $x^2 - \frac{8}{5}x + \frac{2}{5} = 0$. Also, $\frac{r}{s} + \frac{s}{r} = \frac{(r+s)^2 - 2rs}{rs} = \frac{16 - 5}{5/2} = \frac{22}{5} = 4.4$.

16 **[Challenging] • [FRQ] • Target: Inscribed Rectangle in a Parabolic Arch • Cross-Lesson: Topic 8.01 (Area & Perimeter of Inscribed Polygons)**

A rectangle is inscribed under the arch of the parabola $y = 12 - 3x^2$ with its base on the x-axis. If the upper right vertex of the rectangle is at (x, y) with $x = 1$, find the area and perimeter of the rectangle; then determine whether $x = 1$ gives a larger area than $x = \frac{3}{2}$.

By symmetry across the y-axis, for upper-right vertex $(x, 12 - 3x^2)$, the base width is $2x$ and height is $y = 12 - 3x^2$. At $x = 1$, width $= 2$ and height $= 12 - 3 = 9$, so Area $= 18$ and Perimeter $= 2(2 + 9) = 22$. At $x = \frac{3}{2}$, width $= 3$ and height $= 12 - \frac{27}{4} = \frac{21}{4} = 5.25$, giving Area $= 15.75 < 18$.

17 **[Challenging] • [FRQ] • Target: Parametric Line-Parabola Non-Intersection • Cross-Lesson: Topic 2.01 (Compound & Quadratic Inequalities)**

For what range of values of the constant m does the line $y = mx - 5$ NOT intersect the parabola $y = x^2 - 2x + 4$?

Set $x^2 - 2x + 4 = mx - 5 \Rightarrow x^2 - (m + 2)x + 9 = 0$. For no real intersection, the discriminant must be strictly negative: $\Delta = (m + 2)^2 - 4(1)(9) < 0 \Rightarrow (m + 2)^2 < 36 \Rightarrow -6 < m + 2 < 6 \Rightarrow -8 < m < 4$.

18 **[Challenging] • [FRQ] • Target: Pythagorean Theorem & Quadratic Algebraic Identity • Cross-Lesson: Topic 7.01 (Right Triangles & Pythagorean Theorem)**

In right triangle ABC with $\angle C = 90^\circ$, the hypotenuse has length $13\,\text{cm}$ and the sum of the two legs is $17\,\text{cm}$. Set up a quadratic equation to find the lengths of both legs and the area of $\triangle ABC$.

Let one leg be x and the other be $17 - x$. By the Pythagorean theorem, $x^2 + (17 - x)^2 = 13^2 \Rightarrow 2x^2 - 34x + 289 = 169 \Rightarrow 2x^2 - 34x + 120 = 0 \Rightarrow x^2 - 17x + 60 = 0 \Rightarrow (x - 5)(x - 12) =$. The legs are $5\,\text{cm}$ and $12\,\text{cm}$, and the area is $\frac{1}{2}(5)(12) = 30\,\text{cm}^2$.

19 **[Challenging] • [FRQ] • Target: Exponential Equations Reducible to Quadratics • Cross-Lesson: Topic 4.01 (Exponential Functions & Laws of Exponents)**

Solve the exponential-quadratic equation $4^x - 10 \cdot 2^x + 16 = 0$ for all real values of x.

Substitute $u = 2^x > 0$ so that $4^x = (2^x)^2 = u^2$: $u^2 - 10u + 16 = 0 \Rightarrow (u - 2)(u - 8) = 0$. Thus $2^x = 2 \Rightarrow x = 1$, or $2^x = 8 = 2^3 \Rightarrow x = 3$.

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20 [Challenging] • [FRQ] • Target: Structural Parabolic Suspension Modeling • Cross-Lesson: Topic 1.01 (Mathematical Modeling & Evaluation)

A suspension bridge cable hangs in the shape of a parabola $y = ax^2 + c$, where x is the horizontal distance (in meters) from the center of the span and y is the height (in meters) above the road deck. At the center ($x = 0$), the cable is 6 m above the deck; at the towers $x = \pm 60$ m, the cable is 42 m above the deck. Find a and c, and determine the height of a vertical suspender cable located 30 m from the center.

At $x = 0$, $y = c = 6$. At $x = 60$, $42 = a(60)^2 + 6 \Rightarrow 3600a = 36 \Rightarrow a = \frac{1}{100} = 0.01$. Thus $y = 0.01x^2 + 6$. At $x = 30$ m, the height is $y(30) = 0.01(900) + 6 = 15$ m.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Solving Quadratics by Factoring

What is the positive solution to the equation $2x^2 - 5x - 12 = 0$?

a. $\frac{3}{2}$ **b.** 3

c. 4 ✓ **d.** 6

Factor: $(2x + 3)(x - 4) = 0$, which gives $x = -\frac{3}{2}$ or $x = 4$. The positive solution is 4.

22 [Foundational] • [MCQ] • Target: Identifying Vertex Coordinates

In the xy-plane, the parabola $y = -3(x + 4)^2 + 9$ has its vertex at (h, k). What is the value of $h + k$?

a. 5 ✓ **b.** 6

c. 13 **d.** -5

From vertex form $y = a(x - h)^2 + k$, we read $h = -4$ and $k = 9$, so $h + k = -4 + 9 = 5$.

23 [Foundational] • [MCQ] • Target: Equivalent Forms of Quadratic Functions

Which of the following functions is equivalent to $f(x) = x^2 + 8x + 7$?

a. $f(x) = (x + 4)^2 - 9$ ✓ **b.** $f(x) = (x + 4)^2 + 7$

c. $f(x) = (x + 8)^2 - 57$ **d.** $f(x) = (x + 4)^2 - 16$

Complete the square: $x^2 + 8x + 7 = (x^2 + 8x + 16) - 16 + 7 = (x + 4)^2 - 9$.

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24 [Foundational] • [MCQ] • Target: Sum of Solutions via Vieta's Formula

What is the sum of the solutions to the equation $3x^2 - 18x + 7 = 0$?

a. -6

b. $\frac{7}{3}$

c. 6 ✓

d. 18

By Vieta's formula, the sum of the roots of $ax^2 + bx + c = 0$ is $-\frac{b}{a} = -\frac{-18}{3} = 6$.

25 [Foundational] • [MCQ] • Target: Using the Discriminant

How many distinct real solutions does the equation $5x^2 - 4x + 2 = 0$ have?

a. **Zero** ✓

b. Exactly one

c. Exactly two

d. Infinitely many

Compute the discriminant: $\Delta = (-4)^2 - 4(5)(2) = 16 - 40 = -24 < 0$. Since $\Delta < 0$, there are zero real solutions.

26 [Foundational] • [MCQ] • Target: Solving by Square Roots

If $(x - 3)^2 = 20$, what is the greater of the two real solutions for x?

a. $3 + 2\sqrt{5}$ ✓

b. $3 + 4\sqrt{5}$

c. $-3 + 2\sqrt{5}$

d. 23

Take square roots of both sides: $x - 3 = \pm\sqrt{20} = \pm 2\sqrt{5}$, so $x = 3 \pm 2\sqrt{5}$. The greater solution is $3 + 2\sqrt{5}$.

27 [Foundational] • [MCQ] • Target: Graphical Interpretation of a and Δ

The graph of $y = ax^2 + bx + c$ is shown below. Which of the following correctly describes the signs of a and the discriminant $\Delta = b^2 - 4ac$?

a. $a > 0$ and $\Delta > 0$

b. $a > 0$ and $\Delta < 0$

c. $a < 0$ **and** $\Delta > 0$ ✓

d. $a < 0$ and $\Delta = 0$

Because the parabola opens downward, $a < 0$. Because it crosses the x-axis at two distinct points, $\Delta = b^2 - 4ac > 0$.

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28 [Foundational] • [MCQ] • Target: Reconstructing Quadratics from Roots

Which quadratic equation has solutions $x = -5$ and $x = \frac{2}{3}$?

a. $3x^2 + 13x - 10 = 0$ ✓

b. $3x^2 - 13x - 10 = 0$

c. $3x^2 + 17x + 10 = 0$

d. $3x^2 - 17x + 10 = 0$

Reconstruct from factored form: $3(x + 5)\left(x - \frac{2}{3}\right) = (x + 5)(3x - 2) = 3x^2 + 13x - 10 = 0$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Maximum Value of a Quadratic Function

The function $f(x) = -2x^2 + 16x - 25$ reaches its maximum value at $x = a$. What is the maximum value $f(a)$?

a. 4

b. 7 ✓

c. 9

d. 11

The vertex occurs at $a = -\frac{16}{2(-2)} = 4$. Evaluating $f(4) = -2(16) + 16(4) - 25 = -32 + 64 - 25 = 7$.

30 [Intermediate] • [MCQ] • Target: Discriminant Inequalities

In the quadratic equation $kx^2 - 12x + 9 = 0$, k is a nonzero constant. If the equation has no real solutions, which inequality must k satisfy?

a. $k < 4$

b. $k > 4$ ✓

c. $k < -4$

d. $0 < k < 4$

For no real solutions, $\Delta = (-12)^2 - 4(k)(9) = 144 - 36k < 0 \Rightarrow 36k > 144 \Rightarrow k > 4$.

31 [Intermediate] • [MCQ] • Target: Relating Vertex Form to Standard Coefficients

If the parabola $y = x^2 + bx + c$ has vertex $(3, -4)$, what is the value of $b + c$?

a. -1 ✓

b. 1

c. 5

d. 11

Write in vertex form and expand: $y = (x - 3)^2 - 4 = x^2 - 6x + 5$. Thus $b = -6$ and $c = 5$, giving $b + c = -1$.

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32 [Intermediate] • [MCQ] • Target: Axis of Symmetry from Equal Outputs

A quadratic function $g(x)$ satisfies $g(-1) = 8$ and $g(7) = 8$. What is the x-coordinate of the vertex of the graph of $y = g(x)$?

a. 2

b. 3 ✓

c. 4

d. 6

Because $g(-1) = g(7) = 8$, the axis of symmetry lies halfway between $x = -1$ and $x = 7$: $h = \frac{-1+7}{2} = 3$.

33 [Intermediate] • [MCQ] • Target: Product of Real Roots in Reducible Quartics

What is the product of all real solutions to $(x^2 - 5x)^2 - 2(x^2 - 5x) - 24 = 0$?

a. -24 ✓

b. -12

c. 12

d. 24

Let $u = x^2 - 5x$: $u^2 - 2u - 24 = 0 \Rightarrow (u - 6)(u + 4) = 0$. For $x^2 - 5x - 6 = 0$, $\Delta = 25 + 24 = 49 > 0$ (product of two real roots is -6); for $x^2 - 5x + 4 = 0$, $\Delta = 25 - 16 = 9 > 0$ (product of two real roots is 4). Thus all 4 roots are real and their product is $(-6)(4) = -24$.

34 [Intermediate] • [MCQ] • Target: Structure in Quadratic Forms (Vertex vs. Factored)

A ball is thrown upward so that its height h, in feet, after t seconds is $h(t) = -16t^2 + 64t + 80$. Which equivalent form of $h(t)$ displays the time at which the ball hits the ground as a constant or coefficient?

a. $h(t) = -16(t - 2)^2 + 144$

b. $h(t) = -16(t - 5)(t + 1)$ ✓

c. $h(t) = -16t(t - 4) + 80$

d. $h(t) = -8(2t^2 - 8t - 10)$

The ball hits the ground when $h(t) = 0$ ($t > 0$). Factored form $h(t) = -16(t - 5)(t + 1)$ directly displays the positive root $t = 5$ as a constant.

35 [Intermediate] • [MCQ] • Target: Midpoint of a Secant Chord on a Parabola

If the graphs of $y = x^2 - 5x + 7$ and $y = 2x - 3$ intersect at (x_1, y_1) and (x_2, y_2), what is the midpoint of the segment connecting the two intersection points?

a. $(3.5, 4)$ ✓

b. $(3.5, 5)$

c. $(2.5, 2)$

d. $(5, 7)$

Set $x^2 - 5x + 7 = 2x - 3 \Rightarrow x^2 - 7x + 10 = 0 \Rightarrow x_1 + x_2 = 7$. Thus the midpoint has x-coordinate $\bar{x} = \frac{7}{2} = 3.5$, and since it lies on the line $y = 2x - 3$, $\bar{y} = 2(3.5) - 3 = 4$.

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36 [Intermediate] • [MCQ] • Target: Fitting a Quadratic Through Three Points

For all real x, the function $f(x) = ax^2 + bx + c$ satisfies $f(0) = 3$, $f(1) = 2$, and $f(2) = 5$. What is the value of $f(4)$?

a. 19

b. 21

c. 23 ✓

d. 27

Using finite differences: $\Delta f(0) = -1$, $\Delta f(1) = 3$, so the second difference is $2a = 4 \Rightarrow a = 2$. With $c = 3$ and $2(1) + b + 3 = 2 \Rightarrow b = -3$, we have $f(x) = 2x^2 - 3x + 3$, so $f(4) = 2(16) - 12 + 3 = 23$.

37 [Intermediate] • [MCQ] • Target: Interval Length of Quadratic Inequalities

What is the length of the interval of all real x satisfying $x^2 - 4x - 21 \leq 0$?

a. 4

b. 7

c. 10 ✓

d. 14

Factor: $(x - 7)(x + 3) \leq 0$, which holds on $[-3, 7]$. The length of this interval is $7 - (-3) = 10$.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Triangle Area from Parabolic Intercepts • Cross-Lesson: Topic 7.01 (Coordinate Triangle Area)

In the xy-plane, the parabola $y = x^2 - 6x + 8$ intersects the x-axis at A and B and the y-axis at C. What is the area of $\triangle ABC$?

a. 6

b. 8 ✓

c. 12

d. 16

Factoring $x^2 - 6x + 8 = (x - 2)(x - 4)$ gives $A(2, 0)$ and $B(4, 0)$, so the base along the x-axis has length $4 - 2 = 2$. The y-intercept is $C(0, 8)$, giving altitude 8. Thus Area $= \frac{1}{2}(2)(8) = 8$.

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39 [Challenging] • [MCQ] • Target: Cubic Power Sums of Quadratic Roots • Cross-Lesson: Topic 4.01 (Polynomial Identities)

If r and s are the roots of $x^2 - 5x + 3 = 0$, what is the value of $r^3 + s^3$?

a. 80 ✓

b. 95

c. 110

d. 125

We have $r + s = 5$ and $rs = 3$. Using the identity $r^3 + s^3 = (r + s)^3 - 3rs(r + s)$, we obtain $5^3 - 3(3)(5) = 125 - 45 = 80$.

40 [Challenging] • [MCQ] • Target: Radical Equations & Extraneous Roots • Cross-Lesson: Topic 4.01 (Radicals & Rational Exponents)

For how many real values of x is $\sqrt{2x + 15} = x$?

a. Zero

b. **One** ✓

c. Two

d. Three

Square both sides (requiring $x \geq 0$): $2x + 15 = x^2 \Rightarrow x^2 - 2x - 15 = 0 \Rightarrow (x - 5)(x + 3) = 0$. Since $x = -3 < 0$ is extraneous, only $x = 5$ is a valid solution (one real value).

41 [Challenging] • [MCQ] • Target: Circle-Parabola Intersection Trapezoid • Cross-Lesson: Topic 9.01 (Circles) & Topic 8.01 (Trapezoid Area)

A circle with equation $x^2 + y^2 = 13$ and a parabola with equation $y = x^2 - 7$ intersect at four points in the xy-plane. What is the area of the trapezoid formed by these four intersection points?

a. 15

b. 20

c. 25 ✓

d. 30

Substitute $x^2 = y + 7$ into $x^2 + y^2 = 13$: $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$. At $y = 2$, $x^2 = 9 \Rightarrow x = \pm 3$ (top base $b_1 = 6$); at $y = -3$, $x^2 = 4 \Rightarrow x = \pm 2$ (bottom base $b_2 = 4$). The height is $2 - (-3) = 5$, so the trapezoid area is $\frac{6 + 4}{2} \times 5 = 25$.

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42 [Challenging] • [MCQ] • Target: Reconstructing Quadratic Inequalities from Intervals • Cross-Lesson: Topic 2.01 (Inequality Solution Intervals)

If the quadratic inequality $ax^2 + bx + 6 > 0$ has solution set $(-1, 3)$, what is the value of $a + b$?

a. -2

b. 0

c. 2 ✓

d. 4

Since the solution set $(-1, 3)$ is bounded, $a < 0$ and the roots are -1 and 3. Thus $ax^2 + bx + 6 = a(x + 1)(x - 3) = ax^2 - 2ax - 3a$. Matching the constant term gives $-3a = 6 \Rightarrow a = -2$, and $b = -2a = 4$. Therefore $a + b = -2 + 4 = 2$.

43 [Challenging] • [MCQ] • Target: Constrained Optimization on a Closed Interval • Cross-Lesson: Topic 2.01 (Domain Restrictions)

Let $f(x) = x^2 - 4x + 7$. What is the minimum value of $f(x)$ over the restricted interval $3 \le x \le 6$?

a. 3

b. 4 ✓

c. 7

d. 19

Although the global vertex of $f(x) = (x - 2)^2 + 3$ is at $x = 2$ ($f(2) = 3$), $x = 2$ lies outside $[3, 6]$. On $[3, 6]$, $f(x)$ is strictly increasing, so its minimum on $[3, 6]$ occurs at the left endpoint $x = 3$: $f(3) = (3 - 2)^2 + 3 = 4$.

44 [Challenging] • [MCQ] • Target: Arithmetic & Harmonic Means via Quadratics • Cross-Lesson: Topic 6.01 (Statistics & Means)

Two positive numbers have a harmonic mean of 4.8 and an arithmetic mean of 5. What is the positive difference between the two numbers?

a. 1

b. 2 ✓

c. 3

d. 4

Arithmetic mean $\frac{x+y}{2} = 5 \Rightarrow x + y = 10$. Harmonic mean $\frac{2xy}{x+y} = 4.8 \Rightarrow \frac{2xy}{10} = 4.8 \Rightarrow xy = 24$. The numbers are roots of $t^2 - 10t + 24 = 0 \Rightarrow (t - 4)(t - 6) = 0$, so their positive difference is $6 - 4 = 2$.

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45 [Challenging] • [MCQ] • Target: Vertex Distance in the Coordinate Plane • Cross-Lesson: Topic 7.01 (Distance Formula & Pythagorean Triples)

In the xy-plane, what is the shortest distance from the vertex of the parabola $y = 2(x - 3)^2 + 4$ to the origin $(0, 0)$?

a. 5 ✓

b. $\sqrt{13}$

c. 7

d. 25

The vertex of $y = 2(x - 3)^2 + 4$ is $(3, 4)$. Its distance to the origin $(0, 0)$ is $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

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